$$\vec{J}_f = \sigma \vec{E}$$

$$\begin{cases} \vec{\nabla}\cdot\vec{E} = \frac{1}{\varepsilon}\rho_f \\ \vec{\nabla}\cdot\vec{B} = 0 \\ \vec{\nabla}\times\vec{E} = -\frac{\partial\vec{B}}{\partial t} \\ \vec{\nabla}\times\vec{B} = \mu\underbrace{\sigma\vec{E}}_{\vec{J}_f} + \mu\varepsilon\frac{\partial\vec{E}}{\partial t} \end{cases}$$

Continuity eq: $\underbrace{\vec{\nabla}\cdot\vec{J}_f}_{\sigma\vec{\nabla}\cdot\vec{E} = \frac{\sigma}{\varepsilon}\rho_f} + \frac{\partial\rho_f}{\partial t} = 0 \Rightarrow \frac{\partial\rho_f}{\partial t} = -\frac{\sigma}{\varepsilon}\rho_f$

$$\rho_f(t) = e^{-\frac{\sigma}{\varepsilon}t}\rho_f(0)$$

$$\tau = \frac{\varepsilon}{\sigma}$$

Perfect conductor: $\sigma = \infty, \tau = 0 \Rightarrow \rho_f(t) = 0$ (all charge is on the surface)

More realistically, $\tau \ll \frac{1}{\omega}$

$$\Rightarrow \frac{\varepsilon}{\sigma} \ll \frac{1}{\omega} \Rightarrow \frac{\sigma}{\varepsilon\omega} \gg 1$$ for good conductor (metal).

$$\Rightarrow \vec{\nabla}\cdot\vec{E} = \frac{\rho_f}{\varepsilon} \rightarrow \vec{\nabla}\cdot\vec{E} = 0. \text{ (still)}$$

Now $\nabla\times\nabla\times\vec{E} = \nabla(\underbrace{\vec{\nabla}\cdot\vec{E}}_{=0}) - \nabla^2\vec{E} = -\frac{\partial}{\partial t}\nabla\times\vec{B} =$

$$= -\frac{\partial}{\partial t}\left(\mu\sigma\vec{E} + \mu\varepsilon\frac{\partial\vec{E}}{\partial t}\right)$$

$$\Rightarrow \underbrace{\nabla^2\vec{E} = \mu\varepsilon\frac{\partial^2\vec{E}}{\partial t^2}}_{\text{old wave eq.}} + \mu\sigma\frac{\partial\vec{E}}{\partial t}$$

Look for a plane-wave solution

$$\vec{E}(z,t) = \tilde{E}_0 \, e^{i(\tilde{k} z - \omega t)} \quad , \vec{B} = \tilde{B}_0 \, e^{i(\tilde{k} z - \omega t)}$$

$$-\tilde{k}^2 \tilde{E}_0 = \mu\varepsilon(-\omega^2)\tilde{E}_0 + \mu\sigma(-i\omega)\tilde{E}_0$$

$$\tilde{k}^2 = \mu\varepsilon\omega^2 + i\omega\mu\sigma \Rightarrow \tilde{k} = k + i\varkappa$$

$$\tilde{k}^2 = k^2 - \varkappa^2 + 2i\varkappa k = \mu\varepsilon\omega^2 + i\omega\mu\sigma = \mu\omega(\varepsilon\omega + i\sigma)$$

$$\Rightarrow \begin{cases} 2\varkappa k = \omega\mu\sigma \\ k^2 - \varkappa^2 = \mu\varepsilon\omega^2 \end{cases} , \varkappa = \frac{\omega\mu\sigma}{2k}$$

For $\frac{\sigma}{\varepsilon\omega} \gg 1$:

$$\tilde{k}^2 \cong i\sigma\mu\omega$$

$$\tilde{k} = \sqrt{i\sigma\mu\omega} = \frac{1+i}{\sqrt{2}}\sqrt{\sigma\mu\omega}$$

$$k^2 - \frac{\omega^2\mu^2\sigma^2}{4k^2} = \mu\varepsilon\omega^2$$

$$k^4 - \mu\varepsilon\omega^2 k^2 - \left(\frac{\omega\mu\sigma}{2}\right)^2 = 0$$

$$k^2 = \frac{\mu\varepsilon\omega^2 \pm \sqrt{(\mu\varepsilon\omega^2)^2 + 4\left(\frac{\omega\mu\sigma}{2}\right)^2}}{2} =$$

$$= \frac{1}{2}\left(\mu\varepsilon\omega^2 \pm \sqrt{\mu^2\varepsilon^2\omega^4 + \omega^2\mu^2\sigma^2}\right)$$

$$= \frac{1}{2}\mu\varepsilon\omega^2\left(1 \pm \sqrt{1 + \left(\frac{\sigma}{\varepsilon\omega}\right)^2}\right) > 0 \Rightarrow \text{choose } \oplus \text{ sign.}$$

for good metal $\frac{\sigma}{\varepsilon\omega} \gg 1$

$$\begin{cases} k = \omega\sqrt{\frac{\mu\varepsilon}{2}}\left[1 + \sqrt{1 + \left(\frac{\sigma}{\varepsilon\omega}\right)^2}\right]^{1/2} \xrightarrow[\frac{\sigma}{\varepsilon\omega} \gg 1]{} \omega\sqrt{\frac{\mu\varepsilon}{2}}\left[\frac{\sigma}{\varepsilon\omega}\right]^{1/2} = \sqrt{\frac{\mu\sigma\omega}{2}} \\ \varkappa = \frac{\omega\mu\sigma}{2}\frac{1}{k} \Rightarrow \frac{\omega\mu\sigma}{2\sqrt{\frac{\mu\sigma\omega}{2}}} = \sqrt{\frac{\mu\sigma\omega}{2}} = k \end{cases}$$

$$\tilde{k} = k + i\varkappa = \frac{1+i}{\sqrt{2}}\sqrt{\sigma\omega\mu}$$

$$\Rightarrow \vec{E} = \tilde{E}_0\, e^{i(k+i\varkappa)z - i\omega t} = \tilde{E}_0\, e^{-\varkappa z}\, e^{i(kz-\omega t)}$$

El. field <u>decays</u> $e^{-z/\delta}$ where $\delta = \sqrt{\frac{2}{\mu\omega\sigma}} \xrightarrow[\omega\to\infty]{} 0$

No propagation!

<u>Skin depth</u>

Note that $k\delta = 1 \Rightarrow$ decay is very fast, on a wave-length of a wave.

Suppose $\tilde{E}_0 \sim \hat{x} \Rightarrow \tilde{B}_0 \sim \hat{y}$:

$$\vec{\nabla}\times\vec{E} = -\frac{\partial\vec{B}}{\partial t} \Rightarrow i\tilde{k}\times\vec{E} = i\omega\vec{B}$$

$$\vec{B} = \frac{\tilde{k}\times\vec{E}}{\omega} = \frac{1+i}{\sqrt{2}}\frac{\sqrt{\sigma\omega\mu}}{\omega}\,\hat{z}\times\tilde{E}_0 \cdot$$

$$\cdot\, e^{-\varkappa z}\, e^{i(kz-\omega t)}$$

Note $$\frac{|\vec{B}|}{|\vec{E}|} = \sqrt{\frac{\sigma\mu}{\omega}} = \sqrt{\frac{\sigma}{\varepsilon\omega}\,\varepsilon\mu} = \frac{n}{c}\sqrt{\frac{\sigma}{\varepsilon\omega}}$$

$$\varepsilon\mu = \frac{1}{v^2} = \frac{n^2}{c^2}$$

$$\frac{|\vec{B}|}{|\vec{E}|} \sim \frac{\sqrt{\sigma\mu\omega}}{\omega} \sim \frac{1}{\omega\delta} \sim \frac{\mu\sigma}{\tilde{k}^2\delta} \sim \frac{\mu\sigma}{(\tilde{k}\delta)\tilde{k}} \sim \sqrt{\frac{\sigma\mu}{\omega}}$$

but $\omega \sim \frac{\tilde{k}^2}{\mu\sigma}$

$$\sqrt{\frac{\sigma\mu\varepsilon}{\varepsilon\omega}} = \sqrt{\frac{\sigma}{\varepsilon\omega}}\sqrt{\varepsilon\mu} \simeq \frac{n}{c}\sqrt{\frac{\sigma}{\varepsilon\omega}} \gg \frac{n}{c}$$

In metals Magnetic field B enhanced in comparison with electric.

For good metal $\sigma \to \infty \Rightarrow \delta \to 0$

$\Rightarrow \vec{E} \to 0 \Rightarrow$ no el. field inside metal

$\Rightarrow$ complete reflection of EM wave.

Dielectric constant

① Gas of neutral atoms

$$m_e \frac{d^2x}{dt^2} = q E_0 e^{-i\omega t} \quad \underbrace{- m_e \omega_0^2 x}_{\text{restoring force}}$$

$$x(t) = x_0 e^{-i\omega t}$$

$$-x_0 \omega^2 e^{-i\omega t} = \frac{q E_0}{m_e} e^{-i\omega t} - \omega_0^2 x_0 e^{-i\omega t}$$

$$x_0 \left(\omega_0^2 - \omega^2\right) = \frac{q E_0}{m_e}$$

$$x_0 = \frac{q E_0 / m_e}{\omega_0^2 - \omega^2}$$

dipole moment $p = q x_0 = \dfrac{q^2 E_0}{m_e (\omega_0^2 - \omega^2)}$

For a unit volume with n atoms

$$\vec{P} = \frac{n e^2}{m_e (\omega_0^2 - \omega^2)} \vec{E}$$

$$\vec{P} = \varepsilon_0 \chi_e \vec{E} = (\varepsilon - \varepsilon_0) \vec{E}$$

$$\varepsilon = \varepsilon_0 (1 + \chi_e) \Rightarrow \chi_e = \frac{\varepsilon}{\varepsilon_0} - 1$$

$$\varepsilon_0 \chi_e = \varepsilon - \varepsilon_0$$

Thus $$\varepsilon - \varepsilon_0 = \frac{ne^2}{m_e(\omega_0^2-\omega^2)}$$

$$\Rightarrow \varepsilon_r = \frac{\varepsilon}{\varepsilon_0} = 1 + \frac{ne^2}{\varepsilon_0 m_e(\omega_0^2-\omega^2)}$$

For $\omega \ll \omega_0$ (= natural frequency of an atom, i.e. frequency of one of spectral lines)

$$\varepsilon_r = \varepsilon_0\left(1 + \frac{ne^2}{\varepsilon_0 m_e \omega_0^2}\right) > \varepsilon_0$$

$$\Rightarrow n = \sqrt{\frac{\varepsilon\mu}{\varepsilon_0\mu_0}} \approx \sqrt{\frac{\varepsilon}{\varepsilon_0}} > 1 \text{ as well} \Rightarrow V = \frac{c}{n} < c.$$

② Plasma (system of free charges, overall neutral)

$\Downarrow$ also works for metals

set restoring force to $\phi \Rightarrow \omega_0 = 0$.

$$\Rightarrow \varepsilon_r = 1 - \frac{ne^2}{\varepsilon_0 m_e \omega^2} = 1 - \frac{\omega_p^2}{\omega^2}$$

$$\omega_p = \sqrt{\frac{ne^2}{\varepsilon_0 m_e}} = \text{plasma frequency.}$$

But then $\varepsilon_r(\omega > \omega_p) = 1 - \left(\frac{\omega_p}{\omega}\right)^2 < 1$

suggesting $n < 1 \Rightarrow \boxed{V = \frac{c}{n} > c \text{ ?!}}$

Also, for $\omega < \omega_p$ $\varepsilon_r < 0 \rightarrow$ what does it mean?

$$k^2 = \mu\varepsilon\,\omega^2 = \frac{n^2\omega^2}{c^2} \Rightarrow c^2k^2 = \varepsilon_r\,\omega^2 = \left(1-\frac{\omega_p^2}{\omega^2}\right)\omega^2$$

$$\mu\varepsilon = \frac{\mu\varepsilon}{\mu_0\varepsilon_0}\,\mu_0\varepsilon_0 = \frac{n^2}{c^2} \qquad \Rightarrow c^2k^2 = \omega^2-\omega_p^2$$

$$n = \sqrt{\frac{\mu\varepsilon}{\mu_0\varepsilon_0}} \cong \sqrt{\frac{\varepsilon}{\varepsilon_0}} = \sqrt{\varepsilon_r}$$

$$\Rightarrow k = \frac{1}{c}\sqrt{(-1)(\omega_p^2-\omega^2)} =$$

$$= \frac{i}{c}\sqrt{\omega_p^2-\omega^2}$$

$\Rightarrow$ for $\omega < \omega_p$ EM wave <u>attenuates</u>

and decays exponentially with the distance.

But still, what to do about "faster than speed of light" propagation at $\omega > \omega_p$? (no attenuation there!)

This is the case of ~~group~~ <u>phase velocity</u> vs. <u>group velocity</u>

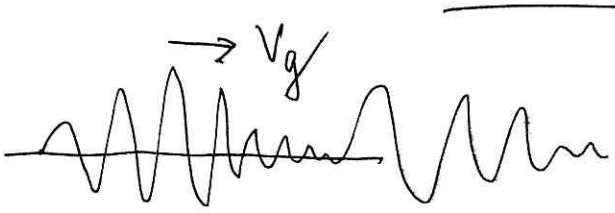
$$V_p = \frac{\omega}{k} = \frac{c}{n} > c \quad \text{for } n<1$$

But <u>theory of relativity</u> does forbid <u>information</u> propagating faster than <u>c</u>.

Can we transport information with <u>infinite plane wave</u>?

NO!

We need to <u>modulate</u> it (so as to transmit pulses $\rightarrow$ encode info in such a Morse code)



$\Rightarrow$ need to consider superposition of plane waves

$$E(x,t) = \int F(k)\, e^{i\varphi(k)}\, dk =$$
$$= \int F(k)\, e^{i(kx - \omega(k)t)}\, dk .$$

From basic Fourier analysis we know that to build a short pulse we need broad range of k

$\Rightarrow$ $F(k)$ is some slow (envelope) function

while $\varphi(k) = kx - \omega(k)t$ varies rapidly with k.

$\Rightarrow$ what matters is the range of <u>least variation of φ</u>

$\Rightarrow$ extremum of $\varphi(k)$: $\dfrac{d\varphi}{dk} = x - \dfrac{d\omega}{dk}t = 0$

$\Rightarrow$ velocity of wave-pulse is given by the <u>group</u> velocity $V_g = \dfrac{d\omega}{dk}$.

Now, for $k^2 = \dfrac{\omega^2 - \omega_p^2}{c^2}$, $\omega^2 = \omega_p^2 + c^2k^2$

$$\cancel{k = \frac{1}{c}\sqrt{\omega^2 - \omega_p^2}} \qquad \omega = \sqrt{\omega_p^2 + c^2k^2}$$

$$\cancel{\Rightarrow \frac{dk}{d\omega}} \qquad \frac{d\omega}{dk} = \frac{2c^2k}{2\sqrt{\omega_p^2 + c^2k^2}}$$

$$\frac{d\omega}{dk} = c\,\frac{\sqrt{\omega^2 - \omega_p^2}}{\omega} = c\sqrt{1 - \frac{\omega_p^2}{\omega^2}}$$

$$\Rightarrow \frac{V_g}{c} = \sqrt{1 - \frac{\omega_p^2}{\omega^2}} < 1 \text{ for } \omega > \omega_p \;!$$

$\Rightarrow$ no problem with special relativity!